Yeast Diauxic Shift Gene Expression

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1. Introduction

In this paper, we investigate gene expression changes in the yeast *Saccharomyces cerevisiae* diauxic shift through metric-based gene selection and K-means clustering, while developing intuition for p-value calculations laying the foundation for Gene Ontology enrichment analysis.

Using DeRisi's data, this project (1) develops and compares custom metrics for identifying variable genes, (2) applies K-means clustering to reveal gene expression patterns, and (3) explores the statistical foundation of Gene Ontology enrichment testing through simulation-based p-value analysis.

2. Methods and Analysis

2.1 Pre-processing

2.1.1 Data

We analyzed two datasets originally from the DeRisi study, provided by Dr. Borislav Hristov, bioinformatic researcher and professor at CalPoly, San Luis Obispo.

1. diauxic_raw_ratios.txt: complete gene expression matrix with 6,152 genes across 7 time points
2. 230_authors.txt: 230 most variable genes as selected by the original authors, representing genes exhibiting substantial expression changes during the shift

2.1.2 Normalization and Transformation

The raw ratio data required log_2 transformation to make fold-changes symmetric, but the 230 genes dataset is already log2 transformed, as evidenced by the data range and presence of both positive and negative values centered around 0.

2.1.3 Handling Duplicate ORF Identifiers

Both datasets included tuples sharing the same Open Reading Frame (ORF) identifier with different expression values.

1. In the authors' dataset, YDR258C appeared twice, so we averaged the expression ratios and kept a single entry.
2. In the raw dataset, grouping by ORF and averaging ratios removed 51 duplicates, reducing the dataset from 6,153 to 6,102 unique genes.

2.1.4 Dataset Alignment

After removing duplicates, the number of unique genes in the authors' selected list dropped from 230 to 228. Therefore, when comparing our most variable genes against the authors', we selected the top 228 from the full dataset.

2.2 Clustering Analysis

Without clustering, gene expression heatmaps lack interpretable structure and fail to group functionally related genes together. Applying clustering algorithms to identify genes with similar expression dynamics, then reordering them based on these clusters, transforms the visualization into biologically interpretable patterns.

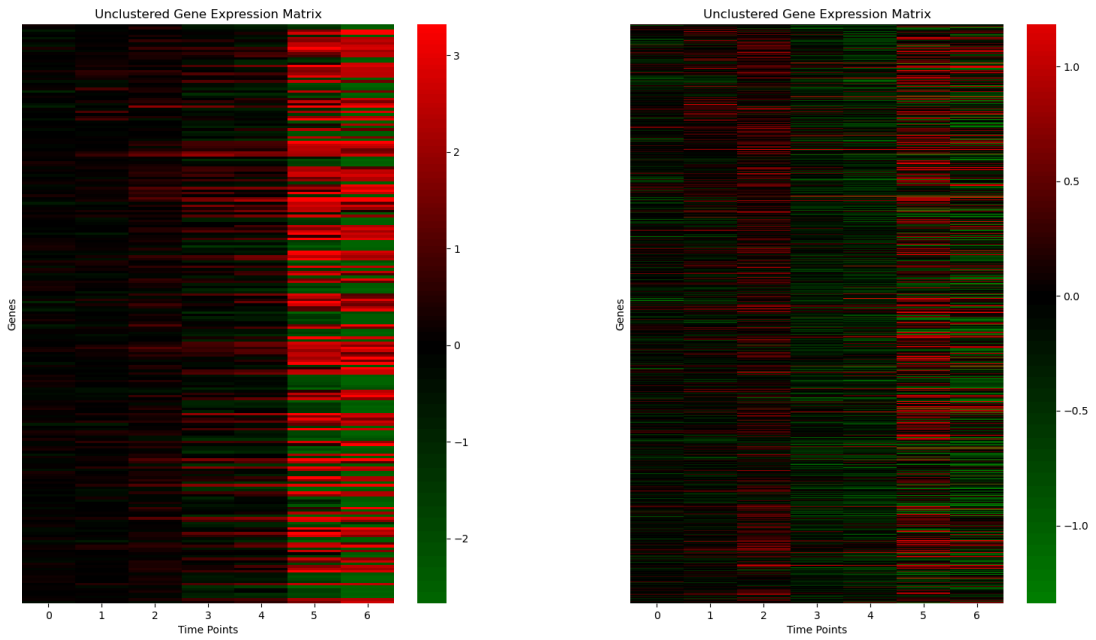

Figure1: Gene expression matrices not showing any specific pattern

2.1.4 Methodology (K-Means Clustering)

The K-mean algorithm requires a pre-defined number of clusters, so to find the optimal number of clusters, we tested cluster counts ranging from 2 to 50 and evaluated each using the silhouette score. To avoid unnecessary computation, we implemented an auto-stop mechanism that halts when the change in silhouette score between consecutive cluster counts falls below a threshold of 0.005. We performed clustering on both datasets.

Authors Curated Gene Set

Best silhouette score: 0.754 (2 clusters)
Auto-stopped at 11 clusters when score changes plateaued
The algorithm selected 2 clusters as optimal

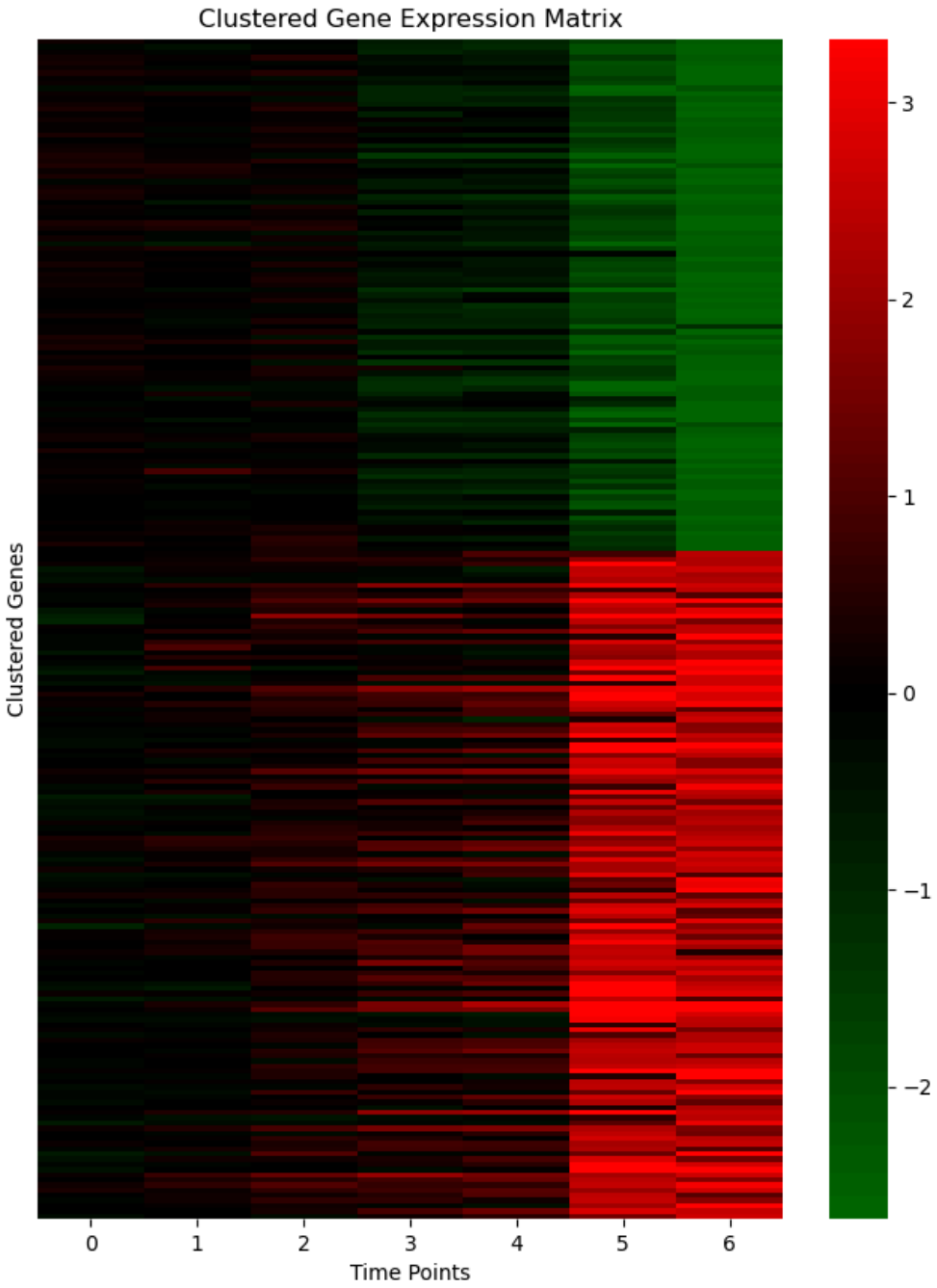

Figure 2: Gene expression matrix showing clear patterns of 2 clusters
Cluster 1: Genes that become down-regulated over time (active during glucose consumption before the shift)
Cluster 2: Genes that become up-regulated (red) as time progresses (getting active for ethanol consumption after the shift)

Clustering the Genome

Best silhouette score: 0.316 (3 clusters)
Auto-stopped at 8 clusters
The algorithm selected 3 clusters as optimal

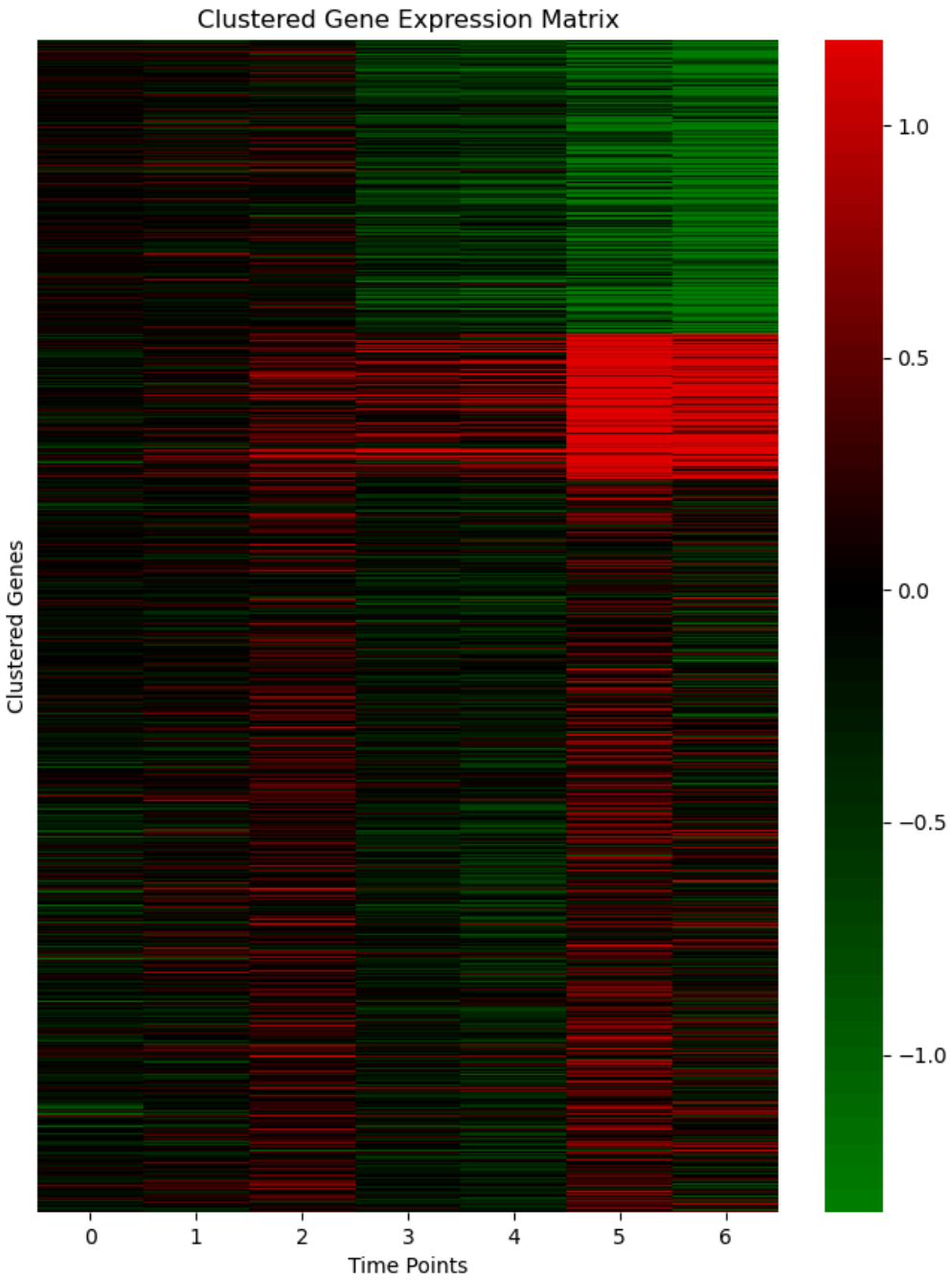

Figure 3: Gene expression matrix of the genome showing some patterns
Cluster 1: very similar to authors' cluster 1 pattern
Cluster 2: very similar to authors' cluster 2 pattern
Cluster 3: Complex alternating pattern (up regulated at t = 2, down regulated at t=3, 4, and up regulated at t=5)

The first two clusters in the patterns are observed in the authors' selection, while the third cluster likely represents genes with expression changes unrelated to the diauxic shift.

2.3 Comparing Selected Most variable genes with Authors

2.3.1 Gene Selection

Focusing on highly variable genes is standard practice in expression analysis. Genes with minimal expression changes or those unrelated to the biological process under investigation reduce statistical power. By selecting the most variable genes, we enrich for genes more likely involved in the diauxic shift.

2.3.2 Variability Metrics

We developed and tested four metrics to quantify gene expression variability:

1. Standard deviation
2. Range
3. Proposed Metric I
4. Proposed Metric II

To evaluate how well we could replicate the authors' gene selection, we tested four metrics for identifying the most variable genes and compared our results using the Jaccard coefficient. Standard deviation measures the overall spread of expression values across time points, treating all timepoints equally. Range calculates the difference between maximum and minimum expression values but is extremely sensitive to outliers. Our first proposed metric computes the absolute difference between mean expression before and after the shift, assuming the transition occurs around timepoint 4. This captures sustained expression but assumes a fixed shift timepoint. Our second proposed metric identifies the maximum absolute deviation from baseline expression at timepoint 0, capturing dramatic responses at any timepoint. However, this metric can be also misled by single extreme measurements, and therefore, it is not able to identify genes with true biological patterns.

Metric	Jaccard Coefficient
Standard Deviation	0.640
Range	0.623
Proposed Metric I	0.515
Propose Metric II	**0.714**

Table 1: Comparison of gene selection metrics against the authors' 228 genes using Jaccard coefficient

2.4 Gene Ontology (GO)

We input 228 selected genes using the Proposed Metric II into the GO enrichment tool. Out of these, 201 genes were identified in the *S. cerevisiae* database, with 193 genes matched. Key biological processes that showed significant over-representation with FDR below 0.05 include:

- Tricarboxylic acid cycle (TCA): 13 genes (FDR of 1.72E-09)
- Aerobic respiration: 18 genes: (FDR of 2.44E-05)

These findings demonstrate that our gene selection is linked to the diauxic shift process. During diauxic shift, yeast cells exhaust glucose and switch to consuming ethanol as their energy source. This switch requires the TCA cycle and aerobic respiration pathways to break down ethanol using oxygen to produce

energy. The strong enrichment of genes in these pathways confirms our selected genes are involved in this metabolic shift.

3. P-value Intuition: Foundation of GO Enrichment

Gene Ontology (GO) enrichment analysis tests whether specific biological functions are over represented in a gene list. We developed simulations across multiple scenarios to build intuition for p-value analysis forming the statistical foundations of GO analysis.

Notation
N := Total population size (total genes in yeast genome)
K := Number of genes in population annotated with a specific GO term (positives in population)
n := Sample size (number of genes in our variable gene list)
k := Observed number of genes in our sample annotated with the GO term (positives in sample)
t := Number of simulation trials

Assume we are sampling 50 genes from the 6,000 genes in the yeast. Of all genes in the genome, 4,260 are annotated with a particular GO term of interest. Of our 50 sampled genes, we observe 38 with this annotation. The question is how rare it is to observe 38 out of 50 genes with this annotation by only chance?

3.1 Empirical P-value Through Simulation

We can estimate the p-value by repeatedly sampling n = 50 genes randomly from the N = 6000 genes population and counting how often we observe >= 38 genes with the annotation. To determine the number of trials needed for accurate estimation, we tested t ∈ {1. 10. …, 1000000} and tracked the change in p-value.

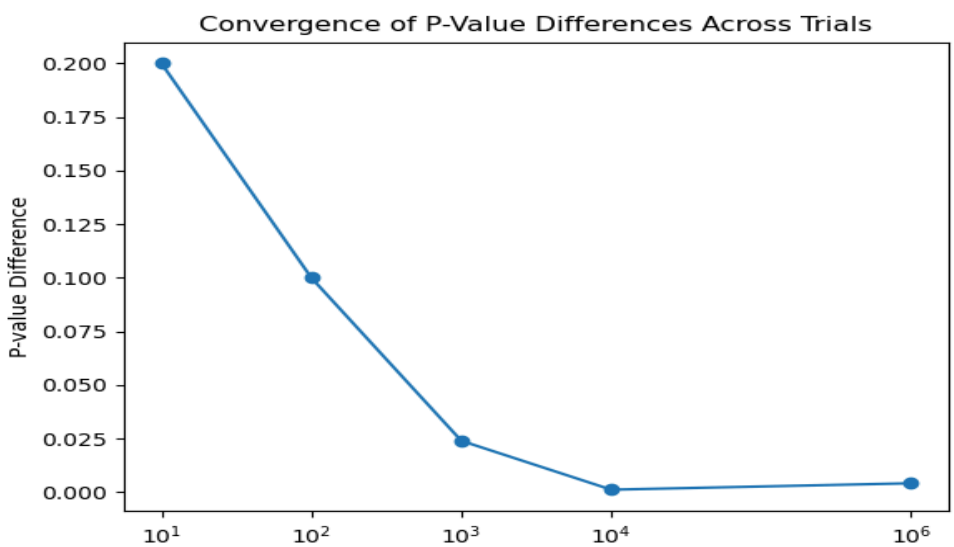

Figure 4: P-value stabilizing by 10,000 trials, with minimal change at 1,000,000 trials

Trial (t)	Empirical p-value	Change from previous
1	0.0	-
10	0.2	0.200
100	0.21	0.00100
1000	0.279	0.0690
10,000	0.271	0.00810
1000,000	0.270	0.000439

Table 2: p-value changes at different trials

We selected t = 1,000,000 for subsequent analysis, balancing precision with computation efficiency.

3.2 Mathematical Modeling: Binomial and Hypergeometric

We can also calculate p-values exactly using probability distributions. When sampling without replacement from a finite population, the hypergeometric distribution models the exact probability of observing a given number of successes. However, when the sample size is small relative to the population, the binomial distribution provides an effective approximation by treating each draw as independent. For our scenario with 6,000 total genes, 4,260 annotated genes, a sample of 50, and 38 observed annotations, all three methods produced very similar results:

- empirical simulation returned p-val of 0.2711
- binomial approximation returned p-val of 0.2713
- hypergeometric calculation returned p-val of 0.2703

This close agreement validates our simulation approach and confirms that the binomial approximation is appropriate when the sample represents a small fraction of the population.

3.3 Statistical Significance of Different Observed Values

To understand how the number of observed annotated genes affects significance, we tested different observed values using $\alpha = 0.05$. Specifically, we calculated p-values for each observed value and compared them to α; observed values with p-values <= α were considered statistically significant.

Observed Value	P-value	Statistically Significant
35	0.630	False
38	0.270	False
40	0.102	False

Table 3: Neither of the observed values is statistically significant given the high prevalence of annotated genes in the population (71%)

3.4 Finding (K, k) Pair for Target P-value Ranges

We searched for (K, k) pairs producing p-values in the range [0.003, 0.01] for our fixed N=6,000 and n=50 and found (K=40, k=3) with the p-value of 0.00433.

(K, k) producing p-value between 0.003 and 0.01 was found by testing K values from 0 to 6,000 in increments of 20, and for each K, testing all k values from 0 to 50. We calculated the hypergeometric p-value for each combination and returned the first pair falling within the target range.

3.5 Multiple Testing Correction

When testing enrichment for each GO term, we perform hundreds of hypothesis tests at the same time, and we expect α of tests to produce p-values < α only by chance. Testing 10000 GO terms, for example, would yield 500 false positives at $\alpha = 0.05$.

A simple correction is the Bonferroni correction, addressing the problem by dividing α by number of tests. For example, for 10 simultaneous tests, α becomes α / 10, reducing false positives and preventing GO terms from being incorrectly identified as over-represented.

4. Implementation Details

Analysis implemented in Python using:

- *numpy*: Matrix operations, data manipulation, and statistical calculations
- *pandas*: Data loading, duplicate handling, and ORF-based grouping
- *scikit-learn*: K-means clustering and silhouette score computation
- *scipy*: Binomial and hypergeometric distribution calculations
- *matplotlib*: Heatmap visualization and convergence plots
- *seaborn*: heatmap styling with custom colormap

Pipeline generates the output files in ./outputs/ directory:

- 228_clustered.png: clustered heatmap of authors' 228 genes
- 228_unclustered.png: unclustered heatmap of authors' 228 genes
- all_genes_clustered.png: clustered heatmap of yeast genome
- all_genes_unclustered.png: unclustered heatmap of yeast genome
- pval_delta_convergence.png: plot showing p-value convergence across trial counts
- my_most_variable_orfs_using_proposed_metric_2.txt: list of 228 most variable genes for GO

K-means clustering analysis used random_state of 13 and n_init of 10. The complete analysis pipeline was performed on an Apple M4 Pro with an estimated run time of 40 seconds. The pipeline can be initiated by running *main.py*, assuming all the dependencies installed.